

Jiayue Wan

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY May 2024

- Ph.D. in Operations Research and Information Engineering, GPA: 4.13/4.3. Advisor: Peter I. Frazier
- Skills: Bayesian Optimization, Statistical Learning, Experimental Design, Causal Inference, Stochastic Modeling, Simulation

Stanford University, School of Engineering, Stanford, CA June 2018

- M.S. in Management Science and Engineering, GPA: 4.07/4.3

Haverford College, Haverford, PA May 2016

- B.S. in Mathematics and Physics, *magna cum laude*, Phi Beta Kappa, GPA: 3.96/4

PROFESSIONAL EXPERIENCE

Susquehanna International Group, Bala Cynwyd, PA August 2024 – current

Quantitative Research Trading Associate

- Developed and implemented mathematical and statistical models to predict price, quote size, market order flow, and other market signals for the index options desk
- Validated signal discoveries through data collection, statistical experiments, and backtesting simulations
- Integrated models into electronic trading strategies, collaborating with developers to optimize performance

Meta, Menlo Park, CA May 2022 – January 2023

Research Engineering Intern, Core Data Science (Adaptive Experimentation)

- Formulated and developed stopping-aware Bayesian optimization algorithms for the BoTorch package for solving expensive-to-evaluate problems such as hyperparameter optimization (HPO) and A/B testing
- Implemented a general model-based learning curve early stopping framework in the adaptive experimentation (Ax) platform

Cornell University, Ithaca, NY April 2020 – May 2022

Data Scientist, COVID-19 Pandemic Response

- Guided Cornell's president and provost on whether to reopen for in-person instruction and what interventions to use, achieving a daily incidence of 0.01% in the 2020-21 academic year among 34K Cornell students/employees
- Developed a Python compartmental simulation model (<https://github.com/peter-i-frazier/group-testing>) to forecast epidemiological outcomes in college environments, whose output influenced policies at Cornell, Stanford, Duke, University of Wisconsin, Boston University, Johns Hopkins, Yale, and several other universities
- Led retrospective parameter estimation and model calibration analysis for the 2020-21 academic year using SQL, Python and Bayesian statistics to support improvements to Cornell's asymptomatic screening program
- Led analysis of the risk of infection during travel to support travel policy decisions and communication with stakeholders by performing causal inference on data from 18K students

Media coverage by Wall Street Journal, Bloomberg, Inside Higher Ed, and Asahi Shimbun.

SELECTED PUBLICATIONS

P.I. Frazier, J.M. Cashore*, N. Duan*, S.G. Henderson*, A. Janmohamed*, B. Liu*, D.B. Shmoys*, **J. Wan***, Y. Zhang*, Modeling for COVID-19 College Reopening Decisions: Cornell, A Case Study. *Proceedings of the National Academy of Sciences*, 119(2) e2112532119 (2022). (*equal contribution)

J. Wan, C.L. Cazer, M.E. Clarkberg, S.G. Henderson, S.E. Lee, G.R. Meredith, M. Osman, D.B. Shmoys, P.I. Frazier, Booster Vaccination Protection Against SARS-CoV-2 Infections in Young Adults During an Omicron BA.1-Predominant Period: A Retrospective Cohort Study. *PLOS Medicine*, 20(1):e1004153 (2023).

P. Buathong*, **J. Wan***, S. Daulton, R. Astudillo, M. Balandat, P.I. Frazier, Bayesian Optimization of Function Networks with Partial Evaluations. *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235:4752-4784 (2024). (*equal contribution)

J. Wan*, Y. Zhang*, P.I. Frazier, Correlation Improves Group Testing: Modeling Concentration-Dependent Test Errors. *Management Science*, Articles in Advance (2025). (*equal contribution)

G.R. Meredith, D.G. Diel, P.I. Frazier, S.G. Henderson, G.A. Koretzky, **J. Wan**, L.D. Warnick, Routine Surveillance and Vaccination on a University Campus During the Spread of the SARS-CoV-2 Omicron Variant. *JAMA Network Open*, 5(5) e2212906 (2022).

RESEARCH EXPERIENCE

COVID-19 Mathematical Modeling

- Formulated a general theoretical framework for correlation in pooled testing to investigate its effect on sensitivity and efficiency and refine the scientific community's understanding of its ability to control epidemics
- Led analysis of vaccine effectiveness in response to queries from the CDC and NYC Health Department using Python and SQL

Grey-Box Bayesian Optimization

- Designed and implemented novel grey-box Bayesian optimization algorithms where additional sources of information besides the final objective are available

TEACHING EXPERIENCE

Teaching Assistant

- Introductory Engineering Probability and Statistics, Cornell University, Fall 2018
- Information Systems and Analysis, Cornell University, Spring 2019
- Simulation Modeling and Analysis, Cornell University, Fall 2019

ACADEMIC SERVICE

Conference Reviewing: AAAI Conference on Artificial Intelligence, International Conference on Automated Machine Learning

Journal Reviewing: Transactions on Machine Learning Research, Nanoscale, Mathematical Biosciences and Engineering, Electronic Research Archive, Data Science in Science

LEADERSHIP & SKILLS

Co-President, Cornell University Operations Research Graduate Students' Association (ORGA) 2020 - 2021

Programming: Python (PyTorch, NumPy, SciPy, Scikit-Learn, Pandas), R, SQL, MATLAB, Julia